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SPIC319
PRODUCT DESIGN AND DEVELOPMENT FOR BIOMEDICAL APPLICATIONS

**SMART KINESIO TAPE WITH MOTION SENSORS AND
LED FEEDBACK FOR REAL TIME KNEE MONITORING**

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INTRODUCTION



- Knee injuries and rehabilitation require proper monitoring
- Incorrect knee movement delays recovery
- Patients performing home exercises lack real-time feedback
- Need for a simple wearable solution for knee motion monitoring



PROBLEM STATEMENT



Conventional
Kinesio tape
provides support but
no motion feedback

Clinical motion
analysis systems are
expensive and non-
portable

Knee braces and
wearables are bulky
and uncomfortable

Lack of real-time
monitoring during
home rehabilitation



AIM



To develop a lightweight and skin-friendly smart Kinesio tape



To provide continuous knee motion tracking using embedded sensors



To monitor knee flexion and extension in real time



OBJECTIVES



Measure knee joint flexion and extension



Design a flexible and wearable sensing system



Enable real-time motion monitoring



Improve rehabilitation monitoring outside clinics



EXISTING SOLUTIONS



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Feature	Existing Product A: Smart Knee Brace	Existing Product B: Motion-Capture IMU	Existing Product C: Fitness Band	Proposed Product: Smart Kinesio Tape
Wearable Format	Bulky brace Yes Limited Medium High	Multi-sensor straps Yes Yes Low Too high	Wristband No Nil High Low-cost	Kinesio tape with tiny module
Angle Tracking				Yes
Real-Time Feedback				Yes
Comfort				High,
Cost				Cost Efficient



LIMITATIONS OF EXISTING SYSTEMS



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No real-time motion feedback

Bulky and rigid form factor

High cost

Not suitable for long-term daily use



GAP IDENTIFIED



No lightweight, flexible, knee-specific motion monitoring solution



Existing systems are expensive and uncomfortable



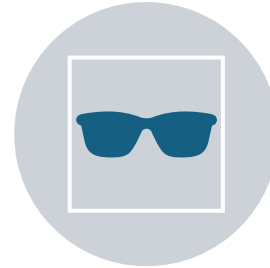
Need for a wearable, affordable, and patient-friendly product



PROPOSED SOLUTION



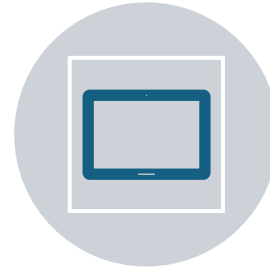
Smart kinesio tape with
embedded motion sensors



Flexible and skin-friendly
design



Real-time knee motion
monitoring



Portable and easy to use



PRODUCT OVERVIEW

Wearable smart Kinesio tape



Tracks knee flexion and extension



Uses IMU sensor and microcontroller



Designed for daily activity and rehabilitation



HARDWARE COMPONENTS



ESP32 Super-Mini C3

MPU6050 IMU Sensor

TP4050 Charging Module

3.7V Lithium-Ion Battery

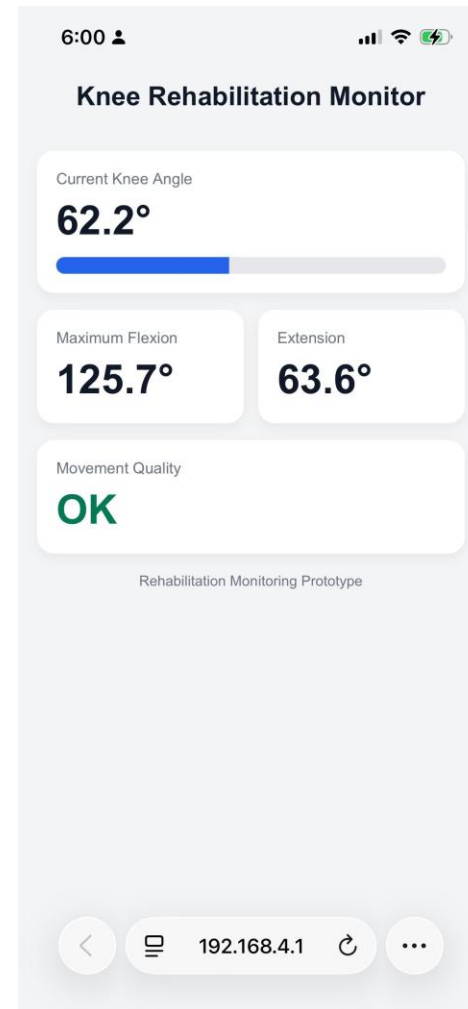
Push Button Switch



WEB UI



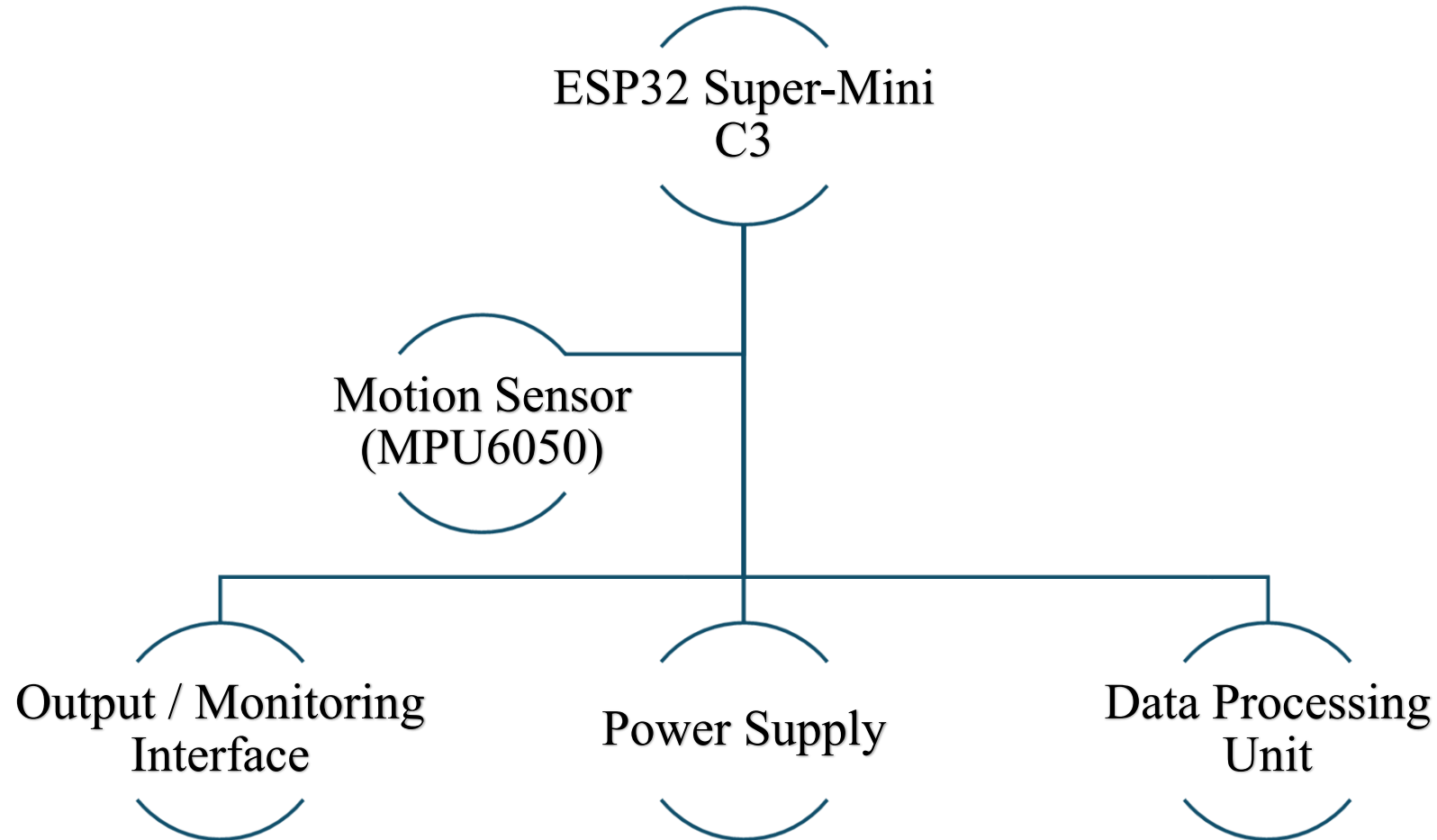
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SYSTEM BLOCK DIAGRAM





WEARABLE DESIGN



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Smart kinesio
tape as wearable
substrate

Flexible and
lightweight

Skin-friendly
and breathable

Does not restrict
knee movement



WORKING PRINCIPLE



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Sensor initializes
and detects knee
movement

Motion data sent to
microcontroller

Knee angle
computed

Motion monitored
continuously

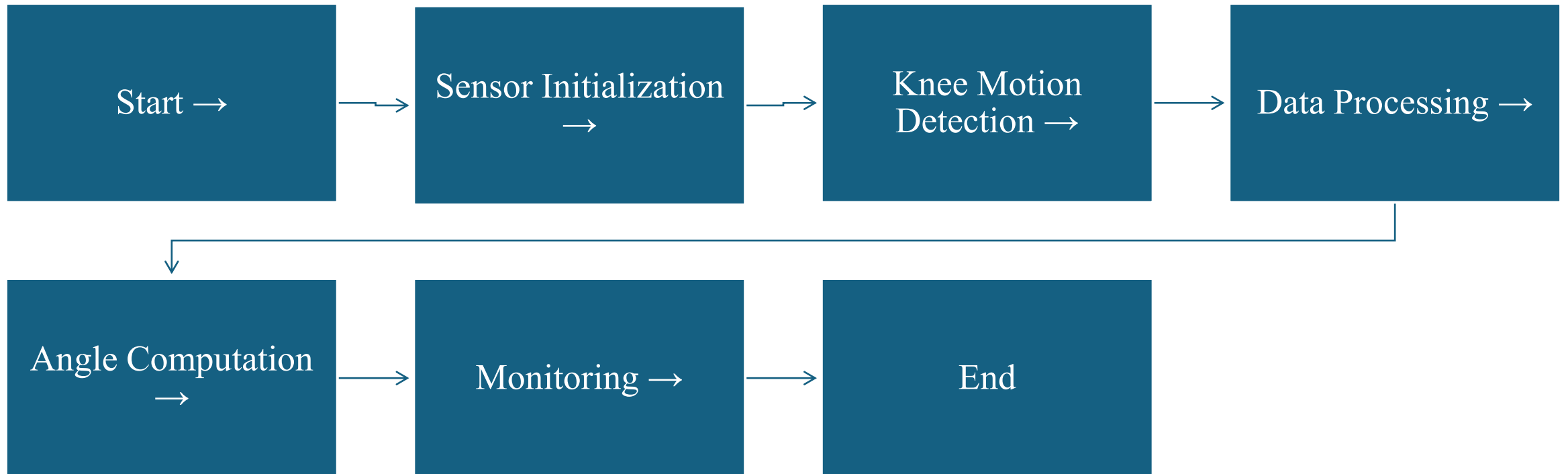
Data logged or
displayed



PROCESSED WORKFLOW



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WEARABILITY TEST



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Device worn during walking and light exercises



Comfort and flexibility assessed



No restriction to natural knee motion



TARGET POPULATION



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Post-surgical knee rehabilitation patients

ACL / PCL injury recovery

Osteoarthritis patients

Athletes recovering from knee injuries



BENEFITS



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For Patients

- Real-time knee monitoring
- Comfortable and lightweight

For Therapists

- Better rehabilitation

For Healthcare

- Cost-effective solution



FEASIBILITY



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Uses commercially available components

Simple wearable fabrication

Low development cost

Scalable for future upgrades



BUDGET



Estimated prototype cost: ₹7,000 – ₹10,000

Lower than existing smart knee devices

Affordable for patients and clinics



TIMELINE



Week 1: Literature review and design



Week 2–3: Prototype development



Week 4: Testing on healthy users



Week 5: Pilot testing



Week 6: Refinement and documentation



FUTURE WORK



Mobile
application
integration



Data analytics
for progress
tracking



Miniaturized
PCB design



Multi-joint
monitoring
support



CONCLUSION



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Smart kinesio tape enables real-time knee motion monitoring

Lightweight and non-invasive solution

Suitable for rehabilitation and daily use

Cost-effective alternative to bulky systems



THANK YOU

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